

Write a program in java that contains the below methods along with the main method. You are free to write extra methods when needed. However, the below ones are mandatory. Your program (source file) should be named as: SectionNameStudentIdAssignmentNumber.java Example: CA1110348Ass3.java. In the Blackboard, you will find an assignment named (Lab Exam). Please upload your solution there.

1. Write a method called **displayMenu()**. You are free to set the return type void/int, it is up to you. This method will be invoked in the main to display the menu. See *sample output*.
2. Write a method defined as **Public static int AaOccurences(String s)**. This method will receive one String from main method, and it will return the number of occurrences of the character 'A' or 'a' in the string variable (s). For example, the method will return 2 if s= "ReALMadrid" as it contains 2 a's. See *sample output*
3. Write a method defined as **public static int SumOfFactorials (int n)**. This method will receive an integer, let's say n. Accordingly, the method will use n to calculate the sum of the factorials. For example, suppose n = 4. The returned value will not be 4!, which is equal to $4 * 3 * 2 * 1$. Instead, it counts $4! + 3! + 2! + 1!$ which represents the integer value returned from the method. See *sample output*.
4. Write a method defined as **public static void KingdomTower(int size)**. This method will receive one integer from main method: the parameter determines the number of rows and number of columns printed. then, the method will print the shape of the Kingdom tower. See *sample output*.
5. The **main** method will do the following:
 - a. Display the menu to allow the user to choose an option.
 - b. If the user chooses option 1, the user will be prompted to enter a String, and this String value will be passed to AaOccurences() method as a parameter.
 - c. If the user entered option 2, the user will be prompted to enter an integer. This integer will be passed to the SumOfFactorials() method as a parameter.
 - d. If the user chooses option 3, the user will be prompted to enter an integer which represents the number of rows and columns -Size- of the figure. If the user entered any odd number greater than or equal 5, the **KingdomTower(size)** method will be invoked, and the integer will be passed to print the figure. If the user entered an odd number -less than 5- or entered an even number, the program should display Error: Enter any odd number greater than 3 (5,7,9,...etc) . see *sample output*.
 - e. If the user chooses option 0, the program will print a final message and will exit.
 - f. The **program should repeat** until the user chooses option 0.



***Sample output 3 pages.

=====

Press 1 for counting the occurrences of letter(A,a) in an input string
Press 2 for calculating the sum of Factorials
Press 3 for displaying the Kingdom Tower
Press 0 to Exit

=====

1
Enter a string
realmAdrid
The letter (A/a) has 2 occurrences in realmAdrid

=====

Press 1 for counting the occurrences of letter(A,a) in an input string
Press 2 for calculating the sum of Factorials
Press 3 for displaying the Kingdom Tower
Press 0 to Exit

=====

2
Enter an integer
4
The sum = 33

=====

Press 1 for counting the occurrences of letter(A,a) in an input string
Press 2 for calculating the sum of Factorials
Press 3 for displaying the Kingdom Tower
Press 0 to Exit

=====

2
Enter an integer
3
The sum = 9



=====

Press 1 for counting the occurrences of letter(A,a) in an input string

Press 2 for calculating the sum of Factorials

Press 3 for displaying the Kingdom Tower

Press 0 to Exit

=====

3

Enter any odd number greater than 3

3

Enter any odd number greater than 3

8

Enter any odd number greater than 3

11

MMMMMMMMMMMM

MM MM

MMM MMM

MMMM MMMM

MMMMMM MMMMM

MMMMMMMMMMMM

MMMMMMMMMMMM

MMMMMMMMMMMM

MMMMMMMMMMMM

MMMMMMMMMMMM

MMMMMMMMMMMM

The tower, with 11 floors, was constructed



=====

Press 1 for counting the occurrences of letter(A,a) in an input string

Press 2 for calculating the sum of Factorials

Press 3 for displaying the Kingdom Tower

Press 0 to Exit

=====

3

Enter any odd number greater than 3

5

MMMMM

MM MM

MMMMM

MMMMM

MMMMM

The tower, with 5 floors, was constructed

=====

Press 1 for counting the occurrences of letter(A,a) in an input string

Press 2 for calculating the sum of Factorials

Press 3 for displaying the Kingdom Tower

Press 0 to Exit

=====

0

BUILD SUCCESSFUL (total time: 42 seconds)



Final Lab Exam